Artilody Watts Trainer

User Manual

Power-based indoor cycling training for Windows, macOS, Linux and Raspberry Pi



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github.com/shenphenchoedron-hue/Artilody-Watts-Trainer

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Contents

1 Introduction . . . 4

1.1 What the app does . . . 4

1.2 Key concepts . . . 4

1.3 System requirements . . . 5

2 Installation . . . 6

2.1 Windows . . . 6

2.2 macOS . . . 6

2.3 Ubuntu 24.04 and Debian 12 . . . 7

2.4 Raspberry Pi OS . . . 7

2.5 Uninstalling . . . 7

2.6 First start . . . 7

3 The main window . . . 8

3.1 Sidebar . . . 9

3.2 Top bar . . . 9

3.3 Window size and scaling . . . 9

4 Setting up your profile . . . 10

4.1 Opening the profile . . . 10

4.2 Profile fields . . . 10

4.3 Training zones . . . 11

4.4 Multiple profiles . . . 12

5 Connecting devices . . . 12

5.1 Scanning . . . 13

5.2 Available devices and connecting . . . 13

5.3 Selected trainer and capabilities . . . 13

5.4 Virtual devices . . . 13

5.5 Connection preferences . . . 14

5.6 Bluetooth diagnostics and connection log . . . 14

5.7 Trainer control (ERG) test . . . 14

5.8 Connection troubleshooting . . . 14

6 Choosing and riding a workout . . . 15

6.1 The workout library . . . 15

6.2 Ride mode: ERG or Free Ride . . . 16

6.3 Starting: the app waits for you to pedal . . . 16

6.4 The Dashboard in detail . . . 16

6.5 Pause, resume and stop . . . 17

6.6 Free ride without a workout . . . 18

6.7 What is recorded . . . 18

7 Workout Builder . . . 19

7.1 Creating and opening workouts . . . 19
7.2 Adding blocks . . . 20
7.3 Interval settings . . . 20
7.4 Timeline chart and block list . . . 20
7.5 Summary, preview and saving . . . 21

8 FTP test . . . 21

8.1 The Ramp FTP Test protocol . . . 21
8.2 How to perform the test . . . 21
8.3 How the FTP is estimated . . . 21
8.4 The “FTP Test Complete” dialog . . . 22
8.5 Using an earlier test from History . . . 22

9 History & Analysis . . . 22

9.1 Period and totals . . . 23
9.2 Selected ride, ride list and charts . . . 23
9.3 Ride details . . . 23
9.4 Exporting CSV . . . 24
9.5 Deleting rides . . . 25

10 Music player . . . 25

10.1 The player card . . . 25
10.2 Music library and playlists . . . 25
10.3 Default playlist . . . 26

11 Settings & data . . . 26

11.1 Settings dialog . . . 26
11.2 Where your data is stored . . . 26
11.3 Backup and restore . . . 27
11.4 Offline and privacy . . . 27

12 Troubleshooting & FAQ . . . 27

13 Glossary & keyboard shortcuts . . . 29

13.1 Glossary . . . 29
13.2 Keyboard shortcuts . . . 29

Appendix A Built-in workouts . . . 30

Appendix B Power zone reference . . . 30

Appendix C License & credits . . . 31

1 Introduction

Artilody Watts Trainer is a free, open-source desktop app for structured, power-based indoor cycling. It connects to your Bluetooth smart trainer and heart-rate sensor, controls the trainer's resistance in ERG mode and records every ride locally on your computer.

1.1 What the app does

- **Rides structured workouts** – eight built-in workouts plus any number of your own, created in the visual Workout Builder.
- **Controls your smart trainer** – in ERG mode the app sends the target power of every interval to the trainer, which sets the resistance so that you produce exactly that power.
- **Shows live data** – power, cadence, heart rate, elapsed time, the workout profile and a timeline of all intervals.
- **Records and analyses** – every second of every ride is stored. History shows totals, charts, time in zones, Normalized Power, Intensity Factor and Training Stress Score, and can export CSV files.
- **Estimates your FTP** – the built-in Ramp FTP Test calculates your Functional Threshold Power and can update your profile and all training zones in one click.
- **Plays music** – a built-in player for your own audio files with playlists and favourites.
- **Works offline** – no account, no cloud, no tracking. All data is kept in a local SQLite database.

1.2 Key concepts

Watts and FTP

Power, measured in **watts (W)**, is the most direct measure of how hard you are working on the bike. Unlike heart rate it responds instantly and is not affected by heat, fatigue or caffeine. **FTP (Functional Threshold Power)** is the highest power you can sustain for roughly one hour. All workouts in Artilody Watts Trainer are stored as a percentage of FTP, so when your FTP changes, every workout target and every training zone is scaled automatically.

ERG mode and Free Ride

In **ERG mode** the app tells the trainer which power to hold. If you pedal faster, the trainer lowers the resistance; if you pedal slower, it increases the resistance – your power stays on target. In **Free Ride** the trainer does not follow the targets: it simulates riding on a flat road, and you control the effort with your gears and cadence. Targets are still shown, so you can follow them by feel. A ride without any workout is also called a Free Ride.

Power zones and RPE

Training intensity is divided into six power zones, each with a colour that is used consistently in charts, the timeline and the dashboard. Each zone also corresponds to a range on the RPE scale (Rating of Perceived Exertion, 1–10), which tells you how hard an effort should feel:

Zone	Name	Default range	RPE	Feels
Z1	Recovery	below 55 % FTP	1	Very easy
Z2	Endurance	55 – 75 % FTP	2 – 3	Easy
Z3	Tempo	75 – 87 % FTP	4 – 6	Moderate
Z4	Sweet Spot	87 – 94 % FTP	7 – 8	Vigorous
Z5	Threshold	94 – 105 % FTP	9	Very hard
Z6	VO_2 Max	above 105 % FTP	10	Max effort

The zone boundaries can be adjusted per profile (see chapter 4). Heart-rate zones are calculated separately with the Karvonen method.

1.3 System requirements

Platform	Requirement
Windows	Windows 10 or 11, 64-bit (x64). Bluetooth LE requires Windows 10 or newer.
macOS	macOS 12 Monterey or newer, Apple Silicon or Intel (universal app).
Linux	Ubuntu 24.04 LTS (amd64) or Debian 12 Bookworm (amd64). The app requires Qt 6.4 or newer, so Ubuntu 22.04 and older are not supported. Bluetooth requires the BlueZ service.
Raspberry Pi	Raspberry Pi OS (64-bit) Bookworm on a Raspberry Pi 3, 4, 5 or 400. The 32-bit edition is not supported. A Pi 4 or 5 is recommended for a smooth UI.
Display	Designed for 1584 × 992 pixels; the whole interface scales down uniformly on smaller windows. Minimum window size 800 × 480 (e.g. the official 7" Raspberry Pi display).
Bluetooth	A Bluetooth Low Energy (BLE, Bluetooth 4.0 or newer) adapter – built-in or USB.

Supported devices

- **Smart trainers** that implement the standard Bluetooth **Fitness Machine Service (FTMS)**. Most current trainers do, for example Tacx NEO, Wahoo KICKR, Saris H3, Elite (Suito, Direto, …) and Zwift Hub. ERG control requires a trainer that supports the FTMS *Set Target Power* command.
- **Heart-rate sensors** that use the standard Bluetooth **Heart Rate Service (HRS)**, for example Polar H10, Garmin HRM-Dual/HRM-Pro or Wahoo TICKR straps and armbands.

> **Warning** **ANT+ is not supported** – only Bluetooth LE. Older trainers that offer only a proprietary Bluetooth protocol (no FTMS) cannot be controlled.

2 Installation

Download the file for your platform from the releases page: https://github.com/shenphenchoedron-hue/Artilody-Watts-Trainer/releases/latest. All packages contain the same version 1.0.0.

Platform	File
Windows 10/11 (64-bit), installer	`ArtilodyWattsTrainer-1.0.0-windows-x64-setup.exe`
Windows 10/11 (64-bit), portable ZIP	`ArtilodyWattsTrainer-1.0.0-windows-x64.zip`
macOS 12 or newer (Apple Silicon and Intel)	`ArtilodyWattsTrainer-1.0.0-macOS-universal.dmg`
Ubuntu 24.04 LTS (amd64)	`artilody-watts-trainer_1.0.0_ubuntu24.04_amd64.deb`
Debian 12 Bookworm (amd64)	`artilody-watts-trainer_1.0.0_debian12_amd64.deb`
Raspberry Pi OS 64-bit Bookworm (Pi 3/4/5/400)	`artilody-watts-trainer_1.0.0_raspberrypi-os-bookworm_arm64.deb`

2.1 Windows

1. Download `ArtilodyWattsTrainer-1.0.0-windows-x64-setup.exe` and run it. The installer creates a Start menu entry and, if you choose, a desktop shortcut.
2. Prefer not to install anything? Extract the ZIP file instead and start `Artilody Watts Trainer.exe` from the extracted folder.
3. **SmartScreen:** the app is not code-signed yet, so Windows may show *“Windows protected your PC”*. Click **More info** → **Run anyway**.
4. Turn on Bluetooth under **Settings** → **Bluetooth & devices**.

2.2 macOS

1. Open the DMG and drag **Artilody Watts Trainer** onto the **Applications** folder.
2. **Gatekeeper:** the app is ad-hoc signed but not notarized by Apple. The first time you open it, right-click (or Control-click) the app and choose **Open**, then confirm with **Open**.
3. If macOS reports that the app is “damaged” or still refuses to open it, run this command once in Terminal:

```
xattr -cr "/Applications/Artilody Watts Trainer.app"
```

Bluetooth permission: when macOS asks whether the app may use Bluetooth, click **Allow**. If you denied access by mistake, enable it again under **System Settings** → **Privacy & Security** → **Bluetooth**. Without this permission the app can only use the virtual devices; the connection log shows *“Bluetooth permission denied”*.

2.3 Ubuntu 24.04 and Debian 12

Download the matching .deb package and install it with apt, which also installs all required Qt 6 libraries:

```
sudo apt install ./artilody-watts-trainer_1.0.0_ubuntu24.04_amd64.deb   # Ubuntu 24.04
sudo apt install ./artilody-watts-trainer_1.0.0_debian12_amd64.deb      # Debian 12
```

Start **Artilody Watts Trainer** from the application menu or run `artilody-watts-trainer` in a terminal. Bluetooth needs the BlueZ service to be running:

```
sudo systemctl enable --now bluetooth
```

2.4 Raspberry Pi OS

You need **Raspberry Pi OS (64-bit) Bookworm** on a Raspberry Pi 3, 4, 5 or 400. Install the arm64 package:

```
sudo apt update
sudo apt install ./artilody-watts-trainer_1.0.0_raspberrypi-os-bookworm_arm64.deb
```

The built-in Bluetooth of the Pi 3/4/5 works with smart trainers and heart-rate sensors. Make sure Bluetooth is switched on in the desktop panel.

> **Tip** On small displays such as the official 7" touch screen (800 × 480), maximise the window. The interface is laid out for 1584 × 992 pixels and scales down uniformly, so every element stays in place – only smaller. A Pi 4 or 5 gives the smoothest charts and animations.

2.5 Uninstalling

- Windows: **Settings** → **Apps** → **Artilody Watts Trainer** → **Uninstall**.
- macOS: move the app from Applications to the Trash.
- Linux and Raspberry Pi: `sudo apt remove artilody-watts-trainer`.

> **Note** Your rides, profiles, workouts and settings are stored separately in your user folder (see section 11.2) and are kept when you uninstall the app.

2.6 First start

On the very first start the app creates its database and fills it with:

- one rider profile (weight 68 kg, FTP 250 W, max HR 175 BPM, resting HR 58 BPM, preferred cadence 85 RPM, default mode ERG) – open the profile and enter your own name and values;
- the eight built-in workouts (see Appendix A);
- a music playlist called *Workout Mix* with the sample tracks shipped with the app;
- a virtual trainer and a virtual heart-rate sensor, so you can try everything without hardware.

A good order for getting started:

1. **Connect your devices** under **Devices** → **Scan for Devices** (chapter 5). Without hardware, connect **Virtual Trainer** and **Virtual HR Sensor**.

2. **Set up your profile** – weight, FTP and heart-rate values (chapter 4). If you don't know your FTP, ride the Ramp FTP Test first (chapter 8).
3. **Choose a workout** under **Workouts**, or build your own in **Builder**.
4. Choose **ERG Mode** or **Free Ride** and press **Start Workout**. Start pedaling – the clock starts automatically.
5. When you are done, press **Stop** and save the ride. Open **History** for the analysis.

3 The main window

The window consists of a navigation sidebar on the left, the current page in the middle and a small top bar in the upper-right corner.



Figure 1. The main window with the Dashboard during a ride: sidebar (left), live metrics and charts (centre), devices, ERG switch, controls and music player (right).

3.1 Sidebar

Item	Purpose
Dashboard	Live view of the current workout or free ride – metrics, charts and controls.
Workouts	The workout library: browse, search, preview and start workouts.
Builder	The Workout Builder for creating and editing workouts.
History	Recorded rides, analysis, time in zones and CSV export.
Devices	Bluetooth scanning, connections, diagnostics and trainer test.
Profile	Opens the User Profile dialog for the current rider.

While a workout is armed or running, a status box appears at the bottom of the sidebar. It shows *Waiting… / Start pedaling*, *Riding* or *Paused* together with the elapsed time and the current power. Click it to jump back to the Dashboard from any page.

3.2 Top bar

- **Sun/moon icon** – switches between the dark and the light theme. The app always starts in the dark theme; the choice applies until you close the app.
- **Gear icon** – opens the Settings dialog (section 11.1).
- **Avatar and name** – opens the profile menu: a list of all profiles (name and FTP) to switch between, **Edit profile…** and **New profile…**. Switching and creating profiles is disabled while a ride is running or paused.

3.3 Window size and scaling

The interface is designed for 1584 × 992 pixels. The window opens at that size (or the available screen size, whichever is smaller). If you make it smaller, the entire interface is scaled down uniformly so the layout is always preserved; it is never scaled above 100 %. The minimum window size is 800 × 480 pixels.

Short messages (for example *“Ride saved – see History for the analysis”*) appear briefly as a notification at the bottom of the window.

4 Setting up your profile

Your profile tells the app how strong you are. FTP drives all power targets and power zones; maximum and resting heart rate drive the heart-rate zones.

4.1 Opening the profile

Click **Profile** in the sidebar, or click your avatar in the top bar and choose **Edit profile…**. The **User Profile** dialog opens.

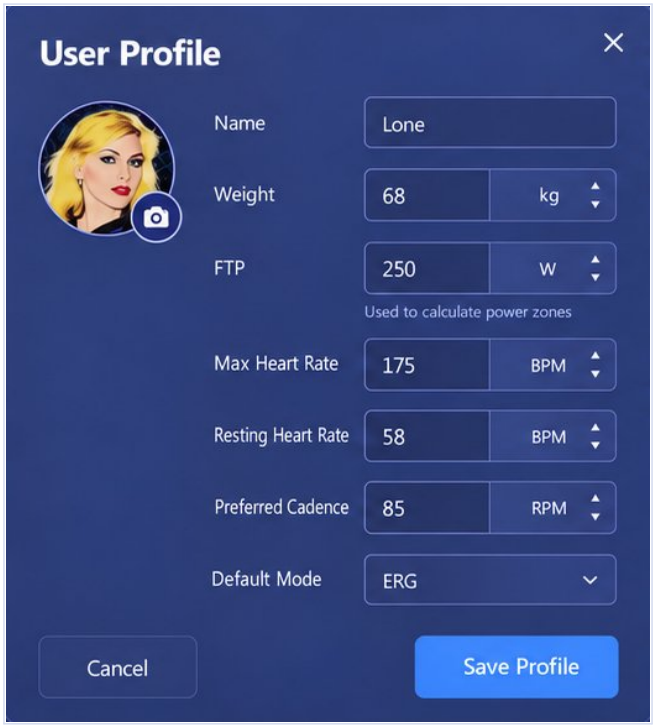


Figure 2. The User Profile dialog (design mockup – the dialog in the app looks very similar). In the app, a “Training zones ›” link below the fields expands the zone editor.

4.2 Profile fields

Field	Range in the dialog	Notes
Name	–	Must not be empty.
Weight	30 – 200 kg, steps of 0.5	Used for W/kg (shown under the FTP field).
FTP	50 – 600 W, steps of 5	Used to calculate power zones and all workout targets.
Max heart rate	100 – 230 BPM	Must be more than 20 BPM above the resting heart rate.
Resting heart rate	30 – 110 BPM	Used together with max HR for the Karvonen zones.
Preferred cadence	50 – 130 RPM	Default cadence target on the dashboard when an interval has no cadence of its own.
Default mode	ERG or Free Ride	Pre-selected ride mode in the workout preview.

Click **Save Profile** to store the changes. If a value is invalid, the dialog stays open and shows the reason in red (for example *“Max heart rate must be above resting heart rate”*). **Cancel** closes the dialog without saving.

Changes to FTP and zones take effect immediately for the workout library, the Builder and the Dashboard. A ride that is already running keeps the profile values it started with.

Avatar

Click the round picture (camera icon) to choose a profile picture (PNG, JPG, JPEG, BMP or WebP). Without a picture the first letter of your name is shown. The app remembers the location of the image file rather than copying it, so keep the file where it is.

4.3 Training zones

Click **Training zones ›** to expand the zone editor on the right side of the dialog.

Power zones

Each of the zones Z1–Z5 has an editable upper boundary in percent of FTP (20–200 %); the next zone starts where the previous one ends, and Z6 covers everything above the Z5 boundary. Next to each zone the resulting watt range for the FTP in the dialog is shown. The boundaries must increase from zone to zone. **Reset** restores the default boundaries 55 / 75 / 87 / 94 / 105 %.

Zone	% of FTP (default)	Watts at FTP 250 W	RPE
Z1 Recovery	0 – 55 %	0 – 137 W	1 Very Easy
Z2 Endurance	55 – 75 %	138 – 187 W	2 – 3 Easy
Z3 Tempo	75 – 87 %	188 – 217 W	4 – 6 Moderate
Z4 Sweet Spot	87 – 94 %	218 – 234 W	7 – 8 Vigorous
Z5 Threshold	94 – 105 %	235 – 262 W	9 Very Hard
Z6 VO_2 Max	above 105 %	263 W and more	10 Max Effort

Heart-rate zones (Karvonen)

Heart-rate zones are calculated from your heart-rate reserve (max HR − resting HR) with the Karvonen formula: *zone limit = resting HR + percentage × (max HR − resting HR)*. They update live while you edit the heart-rate fields and cannot be edited directly.

Zone	% of heart-rate reserve	Example (max 175, rest 58)
Z1 Very Light	50 – 60 %	117 – 128 BPM
Z2 Light	60 – 70 %	128 – 140 BPM
Z3 Moderate	70 – 80 %	140 – 152 BPM
Z4 Hard	80 – 90 %	152 – 163 BPM
Z5 Maximum	90 – 100 %	163 – 175 BPM

A heart rate below 50 % of the reserve is shown as *“Below zone 1”* on the Dashboard.

4.4 Multiple profiles

Several riders can share one installation. Every profile has its own values, zones and ride history; History always shows the rides of the current profile only.

- **Create** – avatar menu → **New profile…**. The dialog starts with an empty name and the values of the current profile; click **Create Profile**. The new profile becomes the current one.
- **Switch** – avatar menu → click a profile name.
- **Delete** – open the profile and click the trash button next to Save Profile. It is only shown when there is more than one profile and no ride is active. Deleting a profile also deletes all of its rides.

> **Note** You cannot switch, create or delete profiles while a ride is running or paused. Stop the ride first. Rides are always stored for the profile that was active when the ride started.

5 Connecting devices

The Devices page finds and connects Bluetooth LE smart trainers and heart-rate sensors, and helps you diagnose connection problems.

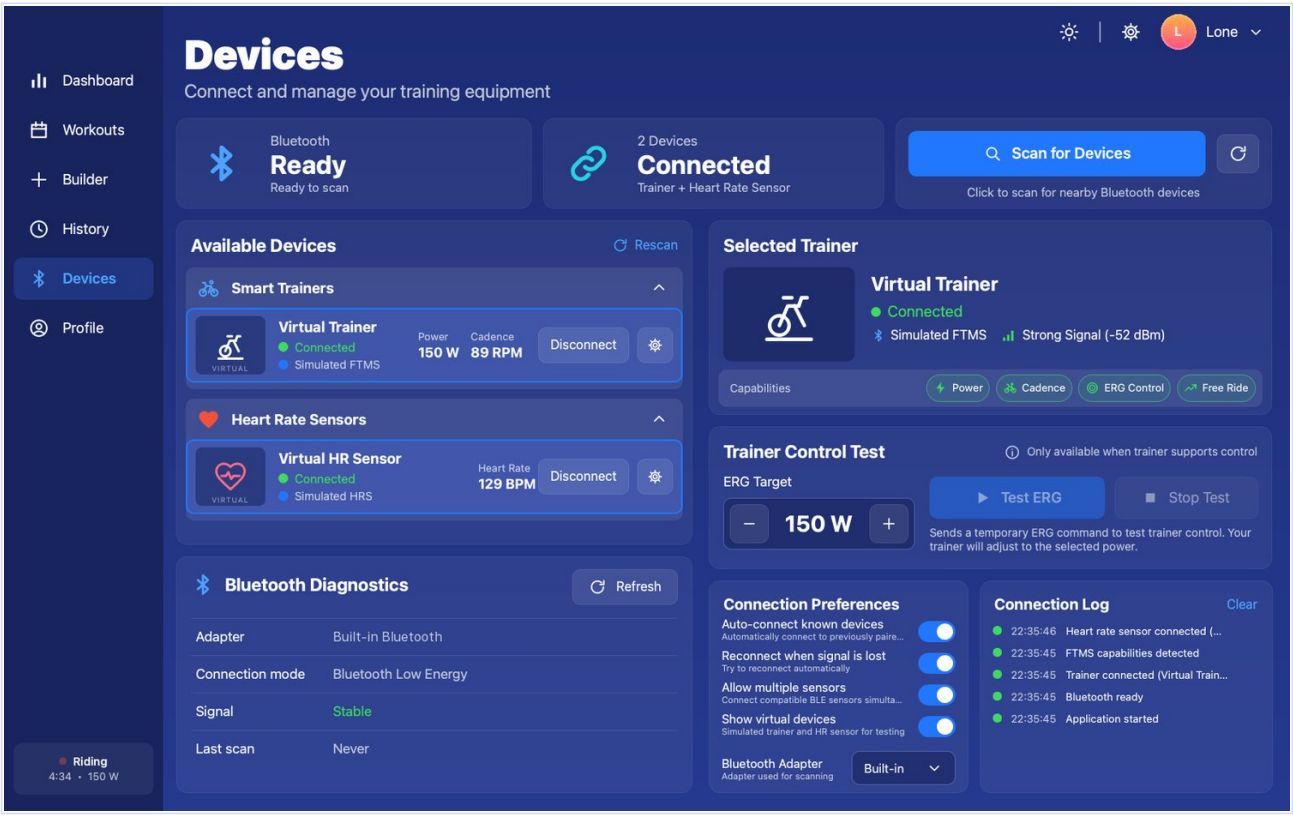


Figure 3. The Devices page with the virtual trainer and virtual heart-rate sensor connected.

5.1 Scanning

Press **Scan for Devices**. The app searches for Bluetooth LE devices for up to 12 seconds; press the button again (**Scanning…**) to stop early. The round arrow button next to it (**Refresh adapter**) and **Rescan** start a new scan. The two status cards at the top show the Bluetooth state (*Ready*, *Off* or *Unavailable*) and how many devices are connected.

Only relevant devices are listed. A device is shown as a **smart trainer** if it advertises the FTMS service, and as a **heart-rate sensor** if it advertises the Heart Rate Service. Devices that do not advertise these services are recognised by typical names (for example KICKR, NEO, Tacx, Saris, Elite, Direto, Suito, Zwift, Wahoo, Magene – or HRM, Polar, H10, TICKR, Coospo, Verity).

> **Tip** Wake your devices before scanning: pedal the trainer for a few turns and moisten/put on the heart-rate strap. Most sensors stop advertising when idle.

5.2 Available devices and connecting

Found devices appear under **Smart Trainers** and **Heart Rate Sensors** (both sections can be collapsed). Each row shows the name, connection state and protocol. Click **Connect** to connect; while connecting the button reads **Cancel**, and once connected **Disconnect**. Connected devices show live values: power and cadence for a trainer, heart rate for a sensor.

The gear button opens the device info: identifier (Bluetooth address), protocol, signal strength (*Strong* ≥ −60 dBm, *Good* ≥ −75 dBm, otherwise *Weak*), status and type.

- A connected device is remembered as your **known** trainer or heart-rate sensor, so it can be connected automatically next time.
- Pressing **Disconnect** forgets the device again, so it will not auto-connect until you connect it manually.
- Only one trainer and one heart-rate sensor can be connected at a time. Connecting another trainer replaces the current one.
- If no heart-rate sensor is connected but your trainer reports heart rate (some trainers pass on a paired strap), the trainer's heart rate is used.

The Dashboard's **Devices** card summarises both connections (*Connected*, *Connecting*, *Offline* or *Not paired*); click it to open the Devices page.

5.3 Selected trainer and capabilities

The **Selected Trainer** card shows the connected trainer, its state, protocol, signal strength and the capabilities detected over FTMS: *Power*, *Cadence*, *ERG Control*, *Free Ride* and *Heart Rate*. If *ERG Control* is missing, the trainer only reports data and cannot be controlled by the app.

5.4 Virtual devices

Virtual Trainer (Simulated FTMS) and **Virtual HR Sensor** (Simulated HRS) are built-in simulators marked with a *VIRTUAL* badge. They let you try every feature – workouts, ERG, History, the FTP dialog – without any hardware. The virtual heart rate reacts to the simulated load and uses the resting and maximum heart rate of your profile. Hide them with **Show virtual devices** (Connection Preferences or Settings). Rides with virtual devices are recorded like any other ride.

5.5 Connection preferences

Option (default: on)	Effect
Auto-connect known devices	At start-up the app connects the remembered trainer and heart-rate sensor. For real devices it starts a scan automatically and connects them as soon as they are found.
Reconnect when signal is lost	If a connection drops, the app retries automatically – after 2 s, then 4 s, 6 s … up to every 30 s. After reconnecting, the last ERG target is sent again.
Allow multiple sensors	Trainer and heart-rate sensor can be connected at the same time. When switched off, only one device can be connected; connecting another device disconnects the first.
Show virtual devices	Lists the virtual trainer and heart-rate sensor.
Bluetooth Adapter	Chooses the adapter used for scanning and connecting: *Built-in* (system default) or a specific adapter if your computer has several (for example a USB dongle).

All preferences are saved in the database and kept between sessions.

5.6 Bluetooth diagnostics and connection log

Bluetooth Diagnostics shows the adapter name, the connection mode (Bluetooth Low Energy), the trainer signal (*Stable* at −75 dBm or better, otherwise *Weak*) and the time of the last scan. **Refresh** starts a new scan.

The **Connection Log** lists time-stamped events with a green (OK) or red (problem) dot – for example *Bluetooth ready*, *Found …*, *FTMS capabilities detected*, *Trainer control granted*, *Reconnecting to … in 4 s* or *Bluetooth permission denied*. **Clear** empties the log. The log is the first place to look when something does not work.

5.7 Trainer control (ERG) test

Use **Trainer Control Test** to check that the app can control your trainer before you start a workout. Set an **ERG Target** with − / + (steps of 10 W, 25–1500 W, default 150 W) and press **Test ERG**. While pedaling you should feel the resistance change; the button shows the measured power. Changing the target during the test sends the new value immediately. **Stop Test** ends the test and returns the trainer to the current workout target, or to free ride if nothing is riding.

> **Note** The test is only available when a trainer is connected and supports control. If the log shows *Trainer control refused*, another app (Zwift, a phone app, a watch) is probably connected to the trainer – close it and try again.

5.8 Connection troubleshooting

- Trainer or sensor not listed: wake it up, move it closer, make sure it is not connected to another app or device (most BLE devices accept only one controlling connection), then scan again.
- Check that Bluetooth is turned on and, on macOS, that the app has Bluetooth permission.
- On Linux and Raspberry Pi check that BlueZ is running: `systemctl status bluetooth`.

- Values drop to 0 or "–": power and cadence are reset after 3 seconds without new data, heart rate after 6 seconds. This usually means a weak signal – reduce the distance or use a USB adapter with an extension cable.
- See chapter 12 for more solutions.

6 Choosing and riding a workout

Pick a workout in the library, choose ERG or Free Ride, and start pedaling – the Dashboard guides you through every interval.

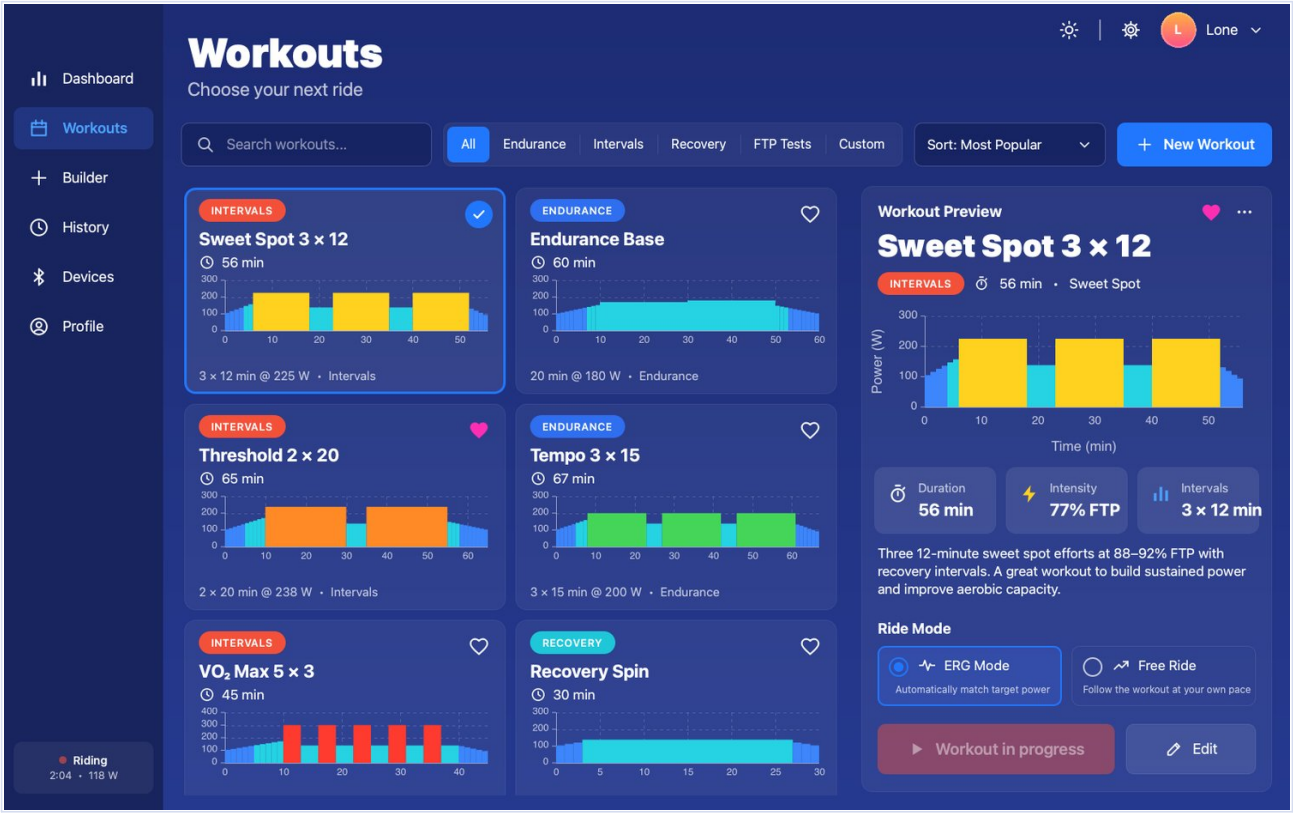


Figure 4. The Workouts library with the preview panel on the right.

6.1 The workout library

- **Search** – filters by name, focus and category as you type.
- **Category** – *All*, *Endurance*, *Intervals*, *Recovery*, *FTP Tests* or *Custom*. Custom shows all your own workouts, whatever their category.
- **Sort** – *Most Popular* (most often started first; default), *Name A–Z*, *Shortest*, *Longest*, *Favorites* (favourites first) or *Newest*.
- **New Workout** – opens an empty workout in the Builder.

Each card shows the category (your own workouts are marked *CUSTOM*), name, duration, a mini profile chart and a summary such as *3 × 12 min @ 225 W • Intervals*. Watt values are always calculated from your current FTP. Click the heart to mark a favourite. Click a card to select it, double-click to open it in the Builder.

Selecting a workout while nothing is riding also loads it on the Dashboard, so you can look at it there before starting.

Workout preview

The preview panel shows the name, category, duration and focus, a larger profile chart, and the tiles **Duration**, **Intensity** (average % of FTP) and **Intervals** (e.g. 3 × 12 min), followed by the description. The ••• (more) menu offers:

- **Duplicate** – creates your own copy named "… (copy)" that you can edit freely.
- **Delete** – removes one of your own workouts after confirmation. Rides already recorded with it stay in History. Built-in workouts cannot be deleted; for the Ramp FTP Test the menu reads *Delete (required for FTP tests)*.

6.2 Ride mode: ERG or Free Ride

Under **Ride Mode** choose **ERG Mode** (*automatically match target power*) or **Free Ride** (*follow the workout at your own pace*). The default comes from your profile. You can also switch ERG on or off during the ride on the Dashboard.

6.3 Starting: the app waits for you to pedal

1. Press **Start Workout**. The app switches to the Dashboard and the workout is **armed**: a blinking *"Start pedaling to begin"* label appears, the Elapsed Time card reads *"Waiting for pedaling"* and the button changes to **Waiting for pedaling…**.
2. While armed, the trainer is left in free-ride mode so you can spin up easily, and readings left over from an earlier ride are cleared.
3. The workout starts automatically as soon as your **cadence reaches 20 RPM** – or, for trainers that do not report cadence, as soon as your **power reaches 30 W**. If you are already pedaling when you press Start Workout, it starts immediately.
4. To start without waiting, press **Start now** in the Dashboard's Controls card.

While a workout is armed you can still choose a different workout and press **Start Workout** again – it replaces the armed one. **Stop** cancels the waiting; nothing is saved.

Ride without workout › (below the Start button) arms a Free Ride without any intervals in the same way.

6.4 The Dashboard in detail

Header

The header shows a breadcrumb (*Indoor Cycling › focus of the workout*), the workout name and a summary line such as *3 × 12 min @ 225 W • Sweet Spot • 56 min total*.

Metric cards

Card	Shows
Power	Current power in watts. The bar is coloured by your current power zone. Footer: the target power and your deviation from it (e.g. *+2 W*).
Cadence	Current cadence in RPM. Footer: the target range ±5 RPM around the interval's cadence, or around your preferred cadence if the interval has none.
Heart Rate	Current heart rate in BPM, or "–" without data. The bar is coloured by heart-rate zone; the footer shows the zone and its range, *Below zone 1* or *No sensor data*.
Elapsed Time	Workout time and the total duration (*of 56:00*).

Workout Profile chart

The chart shows the target power of every interval as bars coloured by RPE zone (legend: *RPE zones 1, 2–3, 4–6, 7–8, 9, 10*), your actual power as it is recorded, and your heart-rate curve in magenta on the right-hand axis. A vertical cursor marks your position, with a label showing your current power. In a Free Ride the time axis grows in 10-minute steps.

Workout Timeline

Below the chart, every interval is shown as a coloured block with its duration, name and target watts (a range for ramps); the current interval is highlighted. The header of the card contains:

- **− 100 % +** – the **intensity** control. Each click makes all targets 5 % easier or harder, from 50 % to 150 %. The adjustment applies immediately (also to the trainer in ERG mode), is recorded in the ride's target power and is reset when you start a new ride.
- **Skip** – jumps to the start of the next interval (only during a ride). The skipped time is not recorded. Skipping the last interval finishes and saves the workout.
- **Edit Workout** – opens the workout in the Builder.
- **Add Interval** – opens the Builder to add blocks (disabled during a ride).

A progress bar with elapsed and total time is shown at the bottom.

Right column

- **Workout Details** – opens a dialog with the description and a list of all intervals with their start time, duration and target.
- **Devices** – connection status of trainer and heart-rate sensor.
- **ERG Mode** switch – turns trainer control on or off at any time during a workout (disabled in a Free Ride). The text below explains the current mode.
- **Controls** – **Start** / **Start now** / **Pause** / **Resume** and **Stop**.
- **Music Player** – see chapter 10.

6.5 Pause, resume and stop

Pause stops the clock and the recording. **While paused, ERG is released** – the trainer returns to free ride so that it does not hold a high resistance while you stand still. **Resume** continues the workout and sends the current target to the trainer again.

Stop opens the *Finish workout?* dialog:

- **Save & Finish** – ends the ride and saves it to History.
- **Discard** – ends the ride without saving.
- **Cancel** – continue riding.

When you reach the end of a workout it is finished and **saved automatically** (*"Workout complete – ride saved to History"*). After a ride, the trainer is always returned to free ride.

> **Warning** Rides shorter than **30 seconds** of recorded time are not saved (*"Ride too short to save"*).

Keyboard shortcut

On the Dashboard, `Space` pauses and resumes the ride. While a workout is waiting for pedaling, `Space` starts it immediately (like Start now); when nothing is armed, it starts the loaded workout immediately. The shortcut is inactive on other pages and while the profile dialog or the FTP result dialog is open.

6.6 Free ride without a workout

Choose **Ride without workout ›** on the Workouts page. The Dashboard shows *Free Ride – ride at your own pace without structured intervals*; there are no targets, ERG is unavailable and the trainer simulates a flat road. Power, cadence and heart rate are still recorded, and the ride is saved under the name *Free Ride*. A free ride runs until you press Stop.

6.7 What is recorded

Once per second the app records your **power**, **cadence**, **heart rate** and the **target power** actually requested (including any intensity adjustment; 0 in a free ride). With the ride it stores the profile, workout name, start time, number of blocks, mode (ERG or Free Ride, as set at the end of the ride) and the FTP at the start. Statistics are calculated with the FTP and zones your profile had when the ride started.

7 Workout Builder

Create your own workouts from blocks, or adapt an existing one. Targets are stored as a percentage of FTP, so your workouts grow with you.

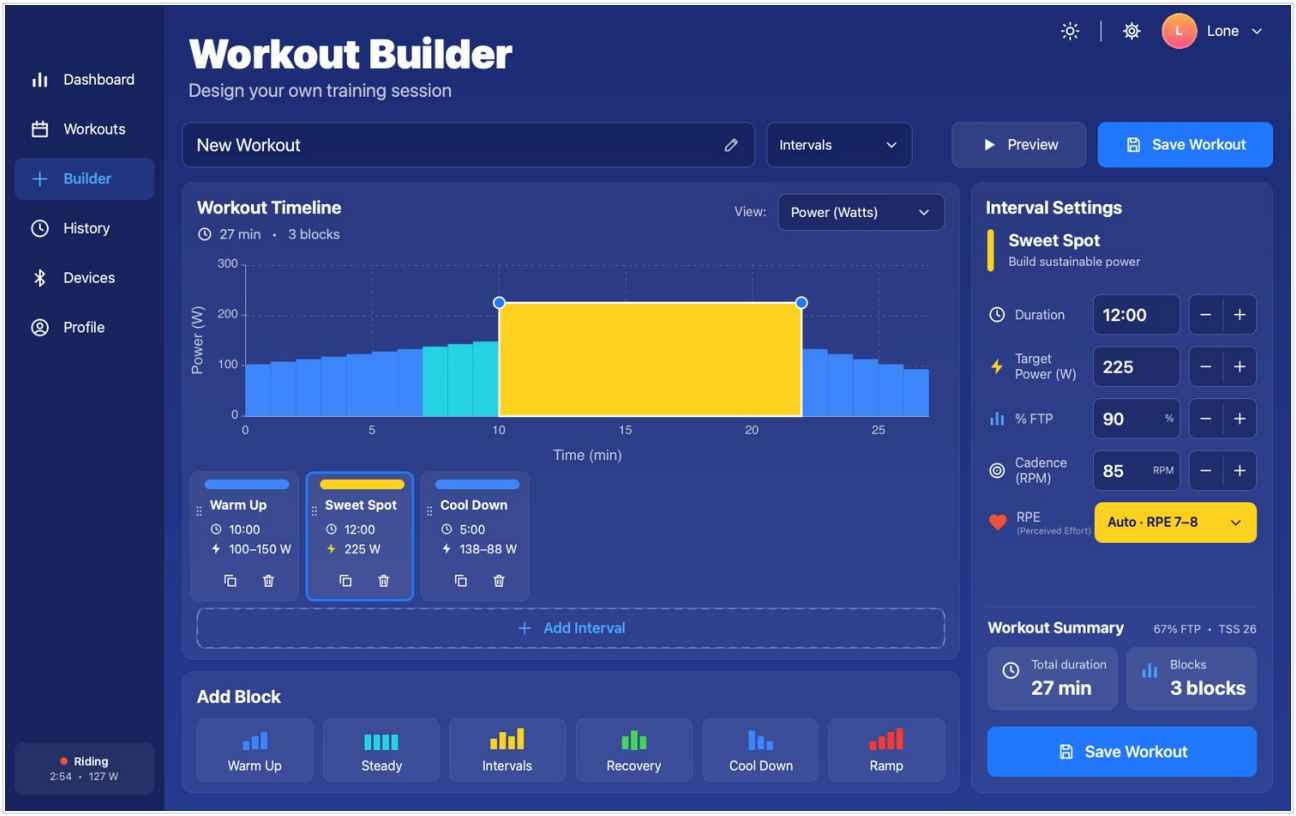


Figure 5. The Workout Builder: timeline chart and block list (left), interval settings and summary (right).

7.1 Creating and opening workouts

- **New workout: Workouts → New Workout**. It starts as *New Workout* (category Intervals) with three blocks: a 10-minute warm-up, a 12-minute sweet-spot interval and a 5-minute cool-down.
- **Edit a workout:** double-click it in the library, choose **Edit** in the preview, or **Edit Workout** on the Dashboard.

At the top of the page you enter the **workout name** and choose the **category** (Endurance, Intervals, Recovery or FTP Tests).

> **Note** A workout in the *FTP Tests* category is treated as an FTP test: when you finish it, the app estimates your FTP (chapter 8).

7.2 Adding blocks

The **Add Block** panel offers six presets. **Add Interval** below the block list adds a Steady block.

Preset	Inserted	Default
Warm Up	always at the start	10:00, ramp 40 → 60 % FTP
Steady	after the selected block	10:00 at 70 % FTP, cadence = your preferred cadence
Intervals	after the selected block	a set of 3 × (5:00 *Interval* at 105 % FTP, cadence 95 + 3:00 *Recovery* at 55 %)
Recovery	after the selected block	5:00 at 55 % FTP
Cool Down	always at the end	5:00, ramp 55 → 35 % FTP
Ramp	after the selected block	10:00, ramp 50 → 100 % FTP

If no block is selected, new blocks are added at the end.

7.3 Interval settings

Select a block in the list or click it in the timeline chart. **Interval Settings** shows its name (click to rename) and these fields, each with − / + buttons and direct input:

Field	Description
Duration	Type *mm:ss* (e.g. 12:00) or *h:mm:ss*; a plain number means minutes. − / + change it by 15 s (up to 2 minutes) or 30 s. Range 5 seconds to 6 hours.
Target Power (W)	Target in watts (steps of 5 W). It is converted to % of your current FTP.
Start / End Power (W)	For ramps, warm-ups and cool-downs: the power at the start and at the end of the block. The target changes linearly between the two.
% FTP	The same target as a percentage of FTP (steps of 1 %, up to 400 %). Watts and % FTP are two views of one value.
Cadence (RPM)	Optional cadence target for the block ("–" = none; the Dashboard then uses your preferred cadence).
RPE	*Auto (from power)* colours the block by the power zone of its target. Choose a fixed RPE (1 Very Easy … 10 Max Effort) to override the colour and zone shown.

> **Tip** A Steady, Recovery or Interval block becomes a ramp only if it is created as one. To build a progressive block, add a Ramp preset and set its start and end power.

7.4 Timeline chart and block list

The **Workout Timeline** chart shows the whole workout with total duration and number of blocks. Use **View:** to switch the vertical axis between **Power (Watts)** and **% FTP**. Click anywhere in the chart to select the block at that time.

In the block list, drag a block by its grip handle to a new position; a marker shows where it will be dropped. The copy button duplicates a block (inserted right after it) and the trash button deletes it.

7.5 Summary, preview and saving

Workout Summary shows the average intensity (% FTP), the estimated TSS, the total duration and the number of blocks. **Preview** opens a large chart of the workout with a **Save Workout** button.

Click **Save Workout** to store the workout (a dot – *Save Workout* • – indicates unsaved changes). A workout needs a name and at least one block. After saving, the workout is selected in the library.

> **Note** Built-in workouts are never overwritten. When you save changes to a built-in workout, the app saves them as a **new personal copy** and leaves the original untouched.

8 FTP test

The built-in Ramp FTP Test estimates your FTP in about 15–25 minutes. Repeat it every 4–8 weeks to keep your zones accurate.

8.1 The Ramp FTP Test protocol

Block	Duration	Target
Warm Up	5:00	40 → 45 % FTP
Ramp	20:00	50 → 150 % FTP, rising continuously by 5 % of FTP per minute

With an FTP of 250 W, the ramp starts at 125 W and would end at 375 W. Almost nobody reaches the end – the test is over when you can no longer hold the target.

8.2 How to perform the test

1. Be rested. Set up a fan, have water ready and make sure the trainer is calibrated if it requires it.
2. Select **Ramp FTP Test** (category *FTP Tests*). **ERG mode** is recommended: the trainer then raises the power for you.
3. Press **Start Workout** and start pedaling.
4. Ride the warm-up easily, then follow the ramp. Keep a steady cadence.
5. When you cannot keep the cadence or the target any longer, press **Stop** and choose **Save & Finish**. If you ride to the end, the ride is saved automatically.

> **Warning** An FTP test is a maximal effort. Do not perform it if you are ill or have health concerns, and consult a doctor if in doubt.

8.3 How the FTP is estimated

The app finds your **best 1-minute average power** during the ride (rolling 60-second window) and calculates:

FTP = 0.75 × best 1-minute power

rounded to whole watts

The result is only offered when it is plausible:

- the ride was saved (at least 30 seconds) and contains at least **5 minutes** of recorded data;
- the estimate lies within the valid FTP range of **50–600 W**;
- the ride belongs to a workout in the *FTP Tests* category;
- the current profile is still the one that rode the test.

8.4 The “FTP Test Complete” dialog

After a successful test the *FTP Test Complete* dialog opens. It shows the **Estimated FTP** (“Based on 75 % of your best 1-minute power (… W)”) and three tiles: **Current FTP**, **Change** (in watts and percent) and **New W/kg**.

- **Update FTP** – saves the new FTP to your profile. Power zones and all workout targets are recalculated immediately (*“FTP updated to … W – zones and workouts adjusted”*).
- **Keep current** – closes the dialog without changes.

Because zones are stored as percentages of FTP, you do not need to edit them after an update.

8.5 Using an earlier test from History

In **History**, select an FTP-test ride and click **View Details**. For FTP tests the details additionally show **Best 1 min** and **Estimated FTP**. Click **Use as FTP** to open the same dialog and apply that result to your profile.

9 History & Analysis

Every saved ride appears in History with a full analysis. Only rides of the current profile are shown.



Figure 6. The History & Analysis page.

9.1 Period and totals

Choose the period in the selector at the top: *Last 7 days*, *Last 30 days* (default), *Last 90 days*, *Last 12 months* or *All time*. Periods count whole days including today. The tiles show **Total Rides**, **Training Time**, **Avg Power** and **Total Energy** (kJ) for that period.

9.2 Selected ride, ride list and charts

- **Selected Workout** – a chart of the selected ride with *Target Power*, *Actual Power* and *Heart Rate*, and a line with date, duration, number of blocks and mode. After a ride, the new ride is selected automatically.
- **Recent Workouts** – date, workout, duration and average power. Click a row to select the ride; **View All** / **Show Less** toggles between recent and all rides in the period.
- **Weekly Training Time** – hours ridden per weekday (Monday–Sunday) within the period.
- **Power Zones** – time and share of time spent in each power zone during the period.
- **Selected Workout Details** – average and maximum heart rate, Normalized Power and average cadence of the selected ride.

9.3 Ride details

View Details opens a dialog with date, start time and duration, and these values:

Value	Meaning
Avg Power / Max Power	Mean and highest one-second power. Seconds without pedaling count as 0 W.
Normalized (NP)	Normalized Power – see below.

Value	Meaning
Energy	Mechanical work in kilojoules (sum of watts per second ÷ 1000). Roughly equal to the food energy burned in kcal.
Intensity (IF)	Intensity Factor = NP ÷ FTP.
TSS	Training Stress Score – see below.
Avg / Max HR	Mean and highest heart rate (seconds without data are ignored).
Avg Cadence	Mean cadence while pedaling (zero cadence is ignored).
Best 1 min, Estimated FTP	Only for FTP-test rides (chapter 8).

A **Time in zones** bar shows how the ride was distributed across the six power zones. Buttons: **Delete ride**, **Use as FTP** (FTP tests only), **Export CSV** and **Close**.

NP, IF and TSS explained

- **Normalized Power (NP)** estimates the power you could have held at a constant output for the same physiological cost. It is calculated from 30-second rolling averages of power, raised to the 4th power, averaged, and the 4th root taken. Rides shorter than 30 seconds use simple average power.
- **Intensity Factor (IF)** = NP ÷ FTP. 0.75 is an endurance ride, about 0.9 a hard interval session, 1.0 equals a one-hour all-out effort.
- **Training Stress Score (TSS)** = duration (s) × NP × IF ÷ (FTP × 3600) × 100. One hour at FTP gives 100 TSS.

All three use the FTP stored with the ride (your FTP when the ride started). The Builder and the library show an *estimated* TSS calculated from the workout targets.

9.4 Exporting CSV

Click **Export CSV** at the top of the page (or in the ride details) and choose:

- **Selected ride (per second)** – suggested file name `ride-YYYYMMDD-HHMM-<workout>.csv`
- **All rides in range (summary)** – suggested file name `watt-history-YYYY-MM-DD.csv`

The save dialog opens in your Documents folder. Files are UTF-8, comma-separated, with a header row.

Per-second file

Three comment lines (`# workout,…`, `# started,…` with ISO date/time, `# ftp,…`) are followed by one row per second:

Column	Content
`seconds`	Seconds since start (0, 1, 2, …)
`elapsed`	Elapsed time as hh:mm:ss
`power_w`	Power in watts
`target_w`	Requested target power in watts (0 in a free ride)
`cadence_rpm`	Cadence in RPM
`heart_rate_bpm`	Heart rate in BPM (0 = no data)

Summary file

One row per ride in the selected period:

Column	Content
`date`	Start date and time (ISO 8601)
`workout`	Workout name
`duration_s`	Recorded duration in seconds
`mode`	ERG or Free Ride
`avg_power_w`, `max_power_w`, `normalized_power_w`	Average, maximum and normalized power
`avg_hr_bpm`, `max_hr_bpm`	Average and maximum heart rate
`avg_cadence_rpm`	Average cadence
`energy_kj`	Energy in kJ
`intensity_factor`	IF (3 decimals)
`tss`	Training Stress Score
`ftp_w`	FTP used for the ride

9.5 Deleting rides

In the ride details, click **Delete ride** and confirm. The ride and all its recorded data are removed permanently. Deleting a workout from the library does *not* delete its rides.

10 Music player

Play your own music while you ride – from the Dashboard, without leaving the app.

10.1 The player card

The **Music Player** card on the Dashboard shows the album art (when the file contains it), title and artist, a heart button, previous / play-pause / next, a position bar, a mute button and a volume slider (default 70 %, remembered). **Up Next** lists the next up to three songs; click one to play it. The playlist icon in the card header opens the music library.

Title and artist are read from the file's tags. Without tags, a file name of the form `Artist - Title.mp3` is split into artist and title; otherwise the file name is used.

10.2 Music library and playlists

- The left column lists your playlists with the number of songs; click one to open it.
- Type a name in **New playlist name** and click **Create**. The trash button deletes the current playlist.
- **Add Files** adds selected audio files; **Add Folder** adds all supported files in a folder and its sub-folders (sorted by name).

- **Shuffle** plays the playlist in random order; **Repeat playlist** (default on) starts again after the last song.
- For each song: play, move up, move down and remove from the playlist.

Favourites: the heart button on the player card adds the current song to a playlist called *Favorites* (created automatically) or removes it again.

Supported formats: MP3, M4A, AAC, FLAC, WAV, OGG, Opus and WMA. Playback uses the operating system's media framework, so whether a format actually plays can depend on the platform (WMA, for example, is typically only available on Windows).

10.3 Default playlist

On first start the app creates the playlist *Workout Mix* and fills it with up to 200 supported files from the first folder that contains music: the sample music shipped with the app (two tracks) or, if none is found, your system Music folder. If you delete the last playlist, an empty *Workout Mix* is created again.

> **Note** Playlists store links to your files – the files are not copied. If you move or delete a file, the player reports *"File not found"*; remove it and add it again from its new location.

11 Settings & data

11.1 Settings dialog

Click the gear icon in the top bar. The Settings dialog contains:

- **Dark theme** – same as the sun/moon button.
- **Show virtual devices** and **Auto-connect known devices** – same as in Connection Preferences on the Devices page.
- The full path of the database file (*"All data is stored locally (offline) in SQLite"*).
- The app version. **Done** closes the dialog.

11.2 Where your data is stored

Everything – profiles, workouts, rides with per-second data, playlists and settings – is stored in a single SQLite file named `watt-trainer.sqlite` in the standard application-data folder of your user account (organisation *Artilody*, application *Watt Trainer*):

System	Default location
Windows	`%APPDATA%\Artilody\Watt Trainer\watt-trainer.sqlite` (usually `C:\Users\<name>\AppData\Roaming\…`)
macOS	`~/Library/Application Support/Artilody/Watt Trainer/watt-trainer.sqlite`
Linux / Raspberry Pi	`~/.local/share/Artilody/Watt Trainer/watt-trainer.sqlite`

The exact path on your computer is always shown in the Settings dialog.

Using a different database file

Set the environment variable `WATT_DB_PATH` to the full path of a database file before starting the app; it will then use (or create) that file instead. Example for macOS/Linux:

```
WATT_DB_PATH="$HOME/watt-test.sqlite" artilody-watts-trainer
```

11.3 Backup and restore

1. Close the app.
2. Copy `watt-trainer.sqlite` from the folder shown in Settings to a safe place (USB drive, cloud folder, another computer).
3. To restore, close the app and copy the backup back to the same folder under the same name.

> **Tip** You can move your complete training history to a new computer the same way. Music files and avatar pictures are referenced by path and must be copied separately.

11.4 Offline and privacy

Artilody Watts Trainer runs fully offline: no account, no cloud, no telemetry or analytics. The app only talks to your Bluetooth devices. Your data leaves your computer only when you export a CSV file or copy the database yourself.

12 Troubleshooting & FAQ

My trainer is not found.

Make sure it is an FTMS (Bluetooth LE) trainer – ANT+-only and proprietary-protocol trainers are not supported. Wake it by pedaling, disconnect it from other apps and devices (Zwift, phone apps, bike computers), check that Bluetooth is on (and, on macOS, that the app has Bluetooth permission), then press Scan for Devices. Look at the Connection Log for errors.

ERG does not change the resistance.

Check that the ERG Mode switch on the Dashboard is on and that the ride is running (ERG is released while waiting, paused and in Free Ride). Check that the trainer lists *ERG Control* under Capabilities, and try the Trainer Control Test. A log entry *Trainer control refused* means another app holds control of the trainer. Some trainers react with a delay of a few seconds.

Heart rate shows “–”.

No heart-rate data has arrived for 6 seconds. Moisten the strap electrodes, make sure the strap is connected on the Devices page, and that it is not paired exclusively to a watch or phone. Replace the battery if the sensor connects but drops out.

Power or cadence drop to 0 for a moment.

Values are reset after 3 seconds without data, typically because of a weak Bluetooth signal. Reduce the distance, avoid metal obstacles and 2.4 GHz interference, or use a USB Bluetooth adapter with an extension cable and select it under Bluetooth Adapter.

The workout does not start.

An armed workout waits until your cadence reaches 20 RPM (or power 30 W). If your trainer reports neither, press Start now or Space.

My ride was not saved.

Rides with less than 30 seconds of recorded time are not saved, and Discard deletes the ride. Rides are saved for the profile that was active when the ride started.

The FTP dialog did not appear after my test.

The estimate is only offered for workouts in the FTP Tests category, with at least 5 minutes of data and a result between 50 and 600 W. You can still open the ride in History → View Details → Use as FTP.

macOS says the app cannot be opened or is damaged.

Right-click the app → Open → Open. If that does not help, run `xattr -cr "/Applications/Artilody Watts Trainer.app"` in Terminal (section 2.2).

Windows shows “Windows protected your PC”.

The app is not code-signed yet. Click More info → Run anyway.

Raspberry Pi: the UI is slow or too small.

Use a Pi 4 or 5 and the 64-bit OS. Maximise the window – the UI scales down to fit and needs at least 800 × 480 pixels. Close other programs while riding.

Music does not play.

Check the volume and mute button on the player card. If a song reports *File not found*, it was moved or deleted; *Cannot play this file* means the format is not supported by your system's media framework – convert it to MP3, M4A or FLAC.

How do I reset all data?

Close the app and delete (or rename, to keep a backup) `watt-trainer.sqlite` in the folder shown in Settings. The next start creates a fresh database with the default profile and the built-in workouts.

Do I need an internet connection?

No. The app works fully offline.

13 Glossary & keyboard shortcuts

13.1 Glossary

Term	Explanation
BLE	Bluetooth Low Energy – the wireless standard used for trainers and sensors.
Cadence	Pedaling rate in revolutions per minute (RPM).
ERG mode	The trainer automatically sets the resistance so you produce the target power, regardless of cadence.
Free Ride	The trainer does not follow power targets; it simulates a flat road and you control the effort. Also a ride without a workout.
FTMS	Fitness Machine Service – the standard Bluetooth profile for smart trainers (data and control).
FTP	Functional Threshold Power – the highest power you can sustain for about one hour.
HRS	Heart Rate Service – the standard Bluetooth profile for heart-rate sensors.
IF	Intensity Factor = Normalized Power ÷ FTP.
Karvonen	Method for heart-rate zones based on the heart-rate reserve (max HR − resting HR).
kJ	Kilojoules of mechanical work; roughly equal to kcal of food energy burned.
NP	Normalized Power – a weighted average power that reflects the physiological cost of a variable ride.
RPE	Rating of Perceived Exertion – how hard an effort feels on a scale of 1 to 10.
TSS	Training Stress Score – training load from duration and intensity; one hour at FTP = 100.
W/kg	Watts per kilogram of body weight – power relative to weight.
Zone	An intensity range defined as a percentage of FTP (power) or heart-rate reserve (HR).

13.2 Keyboard shortcuts

Key	Where	Action
`Space`	Dashboard	Pause / resume the ride. While waiting for pedaling: start now. With nothing armed: start the loaded workout immediately.
`Enter`	New playlist name	Create the playlist.

Appendix A Built-in workouts

These workouts are installed on first start. Watt values depend on your FTP; the examples assume 250 W. Built-in workouts cannot be deleted; edits are saved as a personal copy.

Workout	Category	Duration	Focus / structure
Sweet Spot 3 × 12	Intervals	56 min	Sweet Spot. 6 min warm-up, 3 × 12 min at 90 % FTP (cadence 90) with 5 min recovery at 55 %, 4 min cool-down.
Endurance Base	Endurance	60 min	Build your base. 10 min warm-up, 20 min at 68 % and 20 min at 72 % FTP, 10 min cool-down.
VO_2 Max 5 × 3	Intervals	45 min	VO_2 Max. 10 min warm-up, 5 × 3 min at 120 % FTP (cadence 100) with 3 min recovery, 5 min cool-down.
Recovery Spin	Recovery	30 min	Active recovery. 3 min easy start, 24 min at 55 % FTP, 3 min cool-down.
Threshold 2 × 20	Intervals	65 min	Threshold. 10 min warm-up, 2 × 20 min at 95 % FTP with 5 min recovery, 10 min cool-down.
Ramp FTP Test	FTP Tests	25 min	Progressive ramp to test your FTP. 5 min warm-up, 20 min ramp from 50 to 150 % FTP. Protected – always present.
Tempo 3 × 15	Endurance	67 min	Muscular endurance. 8 min warm-up, 3 × 15 min at 80 % FTP with 4 min recovery, 6 min cool-down.
Over-Unders 4 × 8	Intervals	62 min	Lactate clearance. 10 min warm-up, 4 sets of 8 min alternating 2 min at 92 % and 1 min at 108 % FTP, 4 min recovery between sets, 8 min cool-down.

Appendix B Power zone reference

Zone	Default % FTP	RPE	Typical use
Z1 Recovery	< 55 %	1	Warm-up, cool-down, recovery between intervals and recovery rides.
Z2 Endurance	55 – 75 %	2 – 3	Long aerobic rides; builds the aerobic base.
Z3 Tempo	75 – 87 %	4 – 6	Steady, moderately hard efforts; muscular endurance.
Z4 Sweet Spot	87 – 94 %	7 – 8	Efficient threshold development with manageable fatigue.
Z5 Threshold	94 – 105 %	9	Efforts around FTP, 8–20+ minutes.
Z6 VO_2 Max	> 105 %	10	Short, very hard intervals of 1–5 minutes.

The zone colours are used everywhere in the app: Dashboard power bar, workout charts, timeline blocks, Builder and History. Zone boundaries can be changed per profile (section 4.3).

Appendix C License & credits

Artilody Watts Trainer is free software: you can redistribute it and/or modify it under the terms of the **GNU General Public License version 3** (GPL-3.0) as published by the Free Software Foundation. It is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the `LICENSE` file for details.

- Built with the Qt framework (Qt Quick, Qt Bluetooth, Qt Multimedia, Qt SQL), used under the LGPL/GPL.
- Data storage by SQLite.
- Uses the Bluetooth SIG's open Fitness Machine and Heart Rate service specifications.
- The training metrics NP, IF and TSS follow the widely used definitions by Dr. Andrew Coggan.

Source code, releases and issue tracker: https://github.com/shenphenchoedron-hue/Artilody-Watts-Trainer

Found a bug or have an idea? Please open an issue:
https://github.com/shenphenchoedron-hue/Artilody-Watts-Trainer/issues

This manual describes Artilody Watts Trainer version 1.0.0 (October 2026). Screenshots show the app with the virtual trainer and virtual heart-rate sensor; the profile dialog illustration is a design mockup.